

HXC-108_selftest_evidence

HXC-108 — Video-QA Harness Self-Test Evidence

Harness: `scripts/video_qa/record_feature.sh selftest` **Captured:** 2026-06-22 (UTC), dev host macOS 15.5 **Revision:** 1 **Discipline:** §11.4.6 real evidence · §11.4.3 honest env-gap · §11.4.107(10) self-validated analyzer

What the self-test proves

The self-test proves the harness mechanics on a NON-GUI target (no HXC-112 contention), per the HXC-108 brief: record a deterministic on-screen string, then prove the OCR analyzer READS that known string back (golden-good PASS) AND FAILS on a wrong expected-pattern (golden-bad FAIL). Scope-prefix: `helixcode-harness_selftest-*`.

Result summary

Check	Verdict	Evidence
Project-prefix resolution	PASS	<code>resolve-prefix</code> → <code>helixcode</code> (from <code>.env</code> <code>HELIX_RELEASE_PREFIX</code>)
Analyzer golden-good (expect real string)	PASS (rc=0)	OCR read <code>HELIXCODE_OCR_PROBE_4242</code> , all expected patterns present
Analyzer golden-bad (expect absent string)	FAIL (rc=1)	OCR read real string; <code>HELIXCODE_WRONG_PATTERN_9999</code> correctly NOT matched
Analyzer bluff-catch (string present + TODO implement)	FAIL (rc=1)	anti-bluff pattern <code>todo implement</code> detected → FAIL even though expected pattern present
Analyzer self-validation overall	PASS	golden-good rc=0 AND golden-bad rc=1 → analyzer provably cannot bluff
Live window-scoped recording	SKIP (§11.4.3 env-gap)	both window capture primitives TCC-blocked on host; harness refuses whole-desktop fallback

Self-test exit code: **0** (PARTIAL - PASS: analyzer self-validation PASS; live recording an honest env-gap).

Captured run — analyzer self-validation (the load-bearing anti-bluff proof)

```
[20:18:39] ANALYZER SELF-VALIDATION: PASS (golden-good PASS rc=0, golden-b
```

Captured run — standalone golden-good / golden-bad / bluff-catch

```
--- render a known string into an image (simulating an on-screen frame) --  
frame: PNG image data, 1100 x 220, 8-bit grayscale, non-interlaced
```

```
--- GOLDEN-GOOD: ocr-analyze expects the real string -> MUST PASS ---  
ocr-analyze: image=frame.png extracted_chars=24  
OCR-VERDICT: PASS (all expected patterns present, no bluff): HELIXCODE_OCR  
golden-good exit: 0 (0=PASS expected)
```

```
--- GOLDEN-BAD: ocr-analyze expects a string NOT on screen -> MUST FAIL --  
ocr-analyze: image=frame.png extracted_chars=24  
OCR-VERDICT: FAIL (missing expected pattern(s): HELIXCODE_WRONG_PATTERN_99  
golden-bad exit: 1 (1=FAIL expected)
```

```
--- BLUFF-CATCH: a frame with 'TODO implement' -> MUST FAIL even if expect  
ocr-analyze: image=bluff.png extracted_chars=39  
OCR-VERDICT: FAIL (anti-bluff pattern present: "todo implement")  
bluff-catch exit: 1 (1=FAIL expected)
```

Captured run — live recording honest SKIP (§11.4.3, never faked)

```
[live] launching Terminal window (marker=HELIXCODE_WIN_...) running: echo  
[live] discover: owner+title miss (titles likely hidden by privacy grant);  
[live] discovered window id=27097; capturing window-scoped for 6s  
[live] native window video unavailable (screencapture: capture error The o  
[live] SKIP (honest env-gap §11.4.3): record: window-scoped capture blocke  
      (screencapture -v video AND -l<id> still both failed: 'could not cr  
      Reproduce: screencapture -v -V2 -l27097 /tmp/x.mov ; screencapture  
      Refusing whole-desktop fallback.
```

Host capability probe (re-runnable)

```
scripts/video_qa/record_feature.sh probe-caps:
```

```
host capture capabilities (probed 2026-06-22T20:16:19Z):
```

```
[OK ] screencapture -R region still  
[GAP] screencapture -l<id> window still (id=27097) – window-scoped recor  
[GAP] screencapture -v window video (macOS Screen-Recording video TCC gr  
[OK ] OCR (magick render + tesseract read) under TMPDIR=/Volumes/T7/tmp
```

Root-cause findings (systematic-debugging §11.4.102, fixed during build)

1. **Selftest aborted before analyzer self-validation** — `record_feature` uses `skip()/fail()` which `exit`; a function-level `exit` killed the whole script before the analyzer half. **Fix:** run the live-recording attempt in an isolated subshell (`...`) so its SKIP is contained.
2. **OCR silently read empty under /tmp** — this host's Homebrew tesseract/leptonica 1.87.0 is DENIED read access to `/tmp` (`=/private/tmp`) image files (“`findFileFormat: image file not found`” on a file that exists) but reads files under `$TMPDIR` (`/Volumes/T7/tmp`) and `$HOME` fine. **Fix:** all scratch files moved to a `$TMPDIR`-rooted work dir; never `/tmp`. (This was a hidden FALSE-NEGATIVE bluff surface — OCR would have reported “no content” on perfectly valid frames.)
3. `magick -font Courier` **failed** — the `Courier` font alias is unresolvable on this host. **Fix:** render with a real font file (`/System/Library/Fonts/Menlo.ttc`), default-font fallback otherwise.
4. **Window-title discovery missed** — macOS returns EMPTY window titles to a CLI process without the privacy grant. **Fix:** owner+title first, then documented owner-only fallback (still window-scoped — a single window id — never whole-desktop).
5. **Window capture itself TCC-blocked** — both `screencapture -v` (video) and `-l<id>` (window still) fail on this host. **Outcome:** honest SKIP with reproducer; never a whole-desktop fallback (§11.4.154).

Sources verified

- Self-test + standalone analyzer runs captured live this session — 2026-06-22.